YaoYiwei (Michael) **Cell:13818302572 Email: yaoyiwei2003@126.com**

EDUCATION

2000.9-2004.6	East China Science and Technology University	Mechanical	Bachelor
2012.9-2015.6	East China Normal University	MBA	Master
2018.9-2020.3	Xi'an Jiaotong-liverpool University	Business Analytics	Master(full-time)
2022.8-2023.12	MIT (online)	Statistics and Data Science	Micro Master (Online)
2015.12	PMP Certification		
2021.10-2022.1	Microsoft AZ-900/DP-900 Certified		

Summary

- Data Scientist with 6+ years of experience applying machine learning and AI to industrial and business problems, combined with 14+ years of engineering background in mechanical and manufacturing domains.
- Experienced in building end-to-end AI pipelines spanning image and document understanding, predictive modeling, anomaly detection, and time-series forecasting (ARIMA, LSTM). Strong at translating complex real-world data into scalable, actionable solutions. Promote and mentor AI/Digital tools to internal customers.

WORK EXPERIENCE

2020/09~now BASF China Data Scientist

Machine Learning-Based Predictive Modeling for Manufacturing Parameters

- Partnered with global Coating team to predict optimal equipment settings for improved surface quality.
- Conducted feature engineering, data standardization, outlier removal, and correlation analysis.
- Developed and tuned models (Regression, LightGBM, Ensemble Pipelines) achieving **91% R^2 accuracy**.
- Delivered interactive visualizations comparing "Before vs. After" scenarios to communicate technical results in a clear, business-friendly manner with stakeholders and colleagues from across functions.
- Proposed actionable recommendations, e.g., "Maintain Voltage between 35–45 and keep Flow Rate below 325."

Predictive Maintenance of Reactors

- Developed machine learning models with tagged data from 100+ reactors to predict Remaining Useful Life (RUL).
- Enabled global plants to optimize maintenance schedules, reducing risk and cost.

Time-Series Forecasting for Business & Operational Metrics

- Developed end-to-end time-series forecasting pipelines for business and operational metrics at product-brand level
- Implemented traditional statistical models (e.g. ARIMA) to establish forecasting baselines and interpret temporal patterns
- Extended the pipeline with LSTM-based sequence models to capture long-term dependencies and seasonal effects
- Designed preprocessing workflows including missing-period handling, normalization, and rolling window transformation to support deep learning–based forecasting

Deep Learning-Based Image Processing Solution for Anomaly Detection

- Designed and deployed a **deep learning model using Detectron2** for bubble detection and quality classification.
- Performed data labeling, augmentation, model tuning, inference, and error analysis.

- Achieved **87% accuracy** in bubble detection and deployed the solution via Flask for front-end integration.

LLM-based OCR & Document Processing for Industrial Operations

- Designed and implemented **AI-powered OCR(Florence-2) and document understanding pipelines** to automate quality- and process-related workflows and fine-tuning with LoRA/QLoRA and SFT.
- Extracted structured information from **chemical composition and product specification labels**, including externally available competitor materials.
- Transformed unstructured images and documents into **machine-readable features** for downstream analysis and modeling of **product performance and cost drivers**.
- Integrated information extraction pipelines into internal analytics workflows, improving data consistency and reusability

Generative AI Tasks

- Developed a Generative AI chatbot using RAG with LangChain (chunked documents ingestion) and OpenAI services, enabling teams to quickly extract actionable insights from complex technical documents and improve decision-making efficiency.

2017/06-2018/07 Siemens Healthineer China Senior Mechanical Engineer

- Mechanical design and Failure Model Analysis for X-ray medical product
- Work together with hardware and software team to perform vibration/shock/fall test and troubleshooting on manufacturing.

2014/07-2017/01 Werner Safety Ladder Product China Engineering Manager

- Manage China engineering team to accomplish the mechanical design, testing, supplier management of safety products like ladders and standing platform.

2011/02-2014/07 General Electric China-Transportation Lead Mechanical Engineer

- Mechanical design and test for main structure of control panel used in Transportation.
- Thermal simulation on PCB level includes building a basic model of the PCB assembly with a chip then calculates the consumption and temperature to check if this chip could operate normally within this system.

2010/09-2011/02 Schneider Electric Experienced Mechanical Engineer

- Mechanical design and test for Medium-voltage cabinet.

2010/05-2010/09 Philips China Project Leader

- Product development and supplier management for Vehicle Air Purifier

2007/10-2010/05 General Electric China-Transportation Lead Mechanical Engineer

- Built 2D drawing and 3D model for mechanical design on electronic control panel used in train.

2004/06-2007/09 China Electric Technology Company 50 Institute Mechanical Team Leader

- Mechanical design and test for broadcast communication equipment.